

# Fraud Document Detection Tool

(FDDT)

## Company User Manual

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A guide for client organizations: User (Submitter), Reviewer L1 and Reviewer L2

Describes the application as implemented — companion to the technical documentation in docs/

# Contents

Fraud Document Detection Tool — Guide for Submitting & Reviewing Organizations

Table of Contents

1. System Overview & Architecture
2. Company User Roles & Permissions
3. Signing In & Navigation
4. User (Submitter) Guide
5. Reviewer L1 (Analyst) Guide
6. Reviewer L2 (Senior Reviewer / Supervisor) Guide
7. Getting Help & Administrative Requests

# FDDT Company User Manual

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## Fraud Document Detection Tool — Guide for Submitting & Reviewing Organizations

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Welcome to the **Fraud Document Detection Tool (FDDT)**. This manual provides a complete, step-by-step operational guide for client organizations using FDDT to authenticate supporting documents, detect digital tampering and alterations, and streamline the review process for invoices, reimbursements, quotations, and official records.

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### Table of Contents

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- 1. [System Overview & Architecture](#)
  - 2. [Company User Roles & Permissions](#)
  - 3. [Signing In & Navigation](#)
  - 4. [User \(Submitter\) Guide](#)
    - [Submitting a Single Case](#)
    - [Submitting Cases in Bulk \(Zip Upload\)](#)
    - [Tracking Submissions \(My Cases\)](#)
    - [Understanding the Submitter Case View](#)
  - 5. [Reviewer L1 \(Analyst\) Guide](#)
    - [Operational Reviewer Dashboard](#)
    - [Managing the Review Queue](#)
    - [Full Case Analysis & Risk Breakdown](#)
    - [Forensic & Automated Document Checks](#)
    - [Interactive PDF Visual Overlays](#)
    - [Review Decisions \(Approve, Reject, Escalate\)](#)
    - [Exporting Case Forensic Reports](#)
  - 6. [Reviewer L2 \(Senior Reviewer / Supervisor\) Guide](#)
    - [Managing Escalated Cases](#)
    - [Company Issuer Registry](#)
    - [Company Risk Rules & Thresholds](#)
  - 7. [Getting Help & Administrative Requests](#)
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### 1. System Overview & Architecture

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The **Fraud Document Detection Tool (FDDT)** is an enterprise document verification platform. When third-party supporting documents (such as vendor invoices, school fee receipts, procurement quotations, or travel receipts) are submitted, the automated verification engine performs:

- 1. **Intake Validation:** Checks file format (PDF), verifies integrity, ensures the file is not password-protected, and validates file size limits.
- 2. **Text Extraction & Normalization:** Automatically reads all printed text (including multi-lingual content such as Arabic and English) and structures key fields (dates, amounts, invoice numbers, vendor names).

3. **Forensic Analysis:** Scans for copy-move cloning, digital alteration artifacts (Error Level Analysis), text deleted or rewritten in converted scans (ghost text), retyped text in a different font or a second copy of the font, suspicious PDF metadata histories (including documents built as web pages), and visual inconsistencies.
4. **Consistency & Verification:** Validates calculations, compares cross-document values within the case, checks the vendor against your company's known **Issuer Registry**, and performs signature/stamp detection.
5. **Explainable Risk Scoring:** Aggregates findings into a transparent risk score (0–100) mapped to risk tiers (**Low, Medium, High**) with clear explanations.

**Tenant Data Privacy:** Your organization operates in an isolated workspace. Users within your company only ever see cases, documents, and settings belonging to your company.

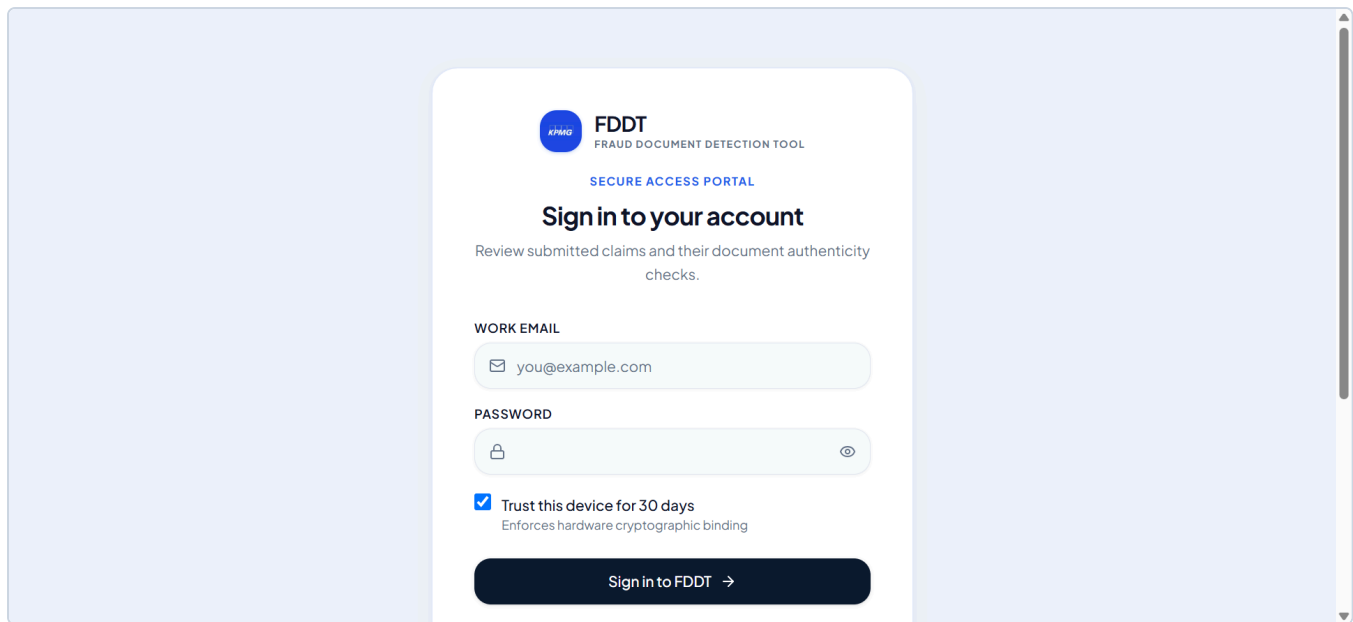
## 2. Company User Roles & Permissions

Within your company, three distinct user roles exist to enforce separation of duties:

| Role                                          | Primary Responsibility                  | Available Screens & Capabilities                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|-----------------------------------------------|-----------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>User</b> ( <i>Submitter</i> )              | Document Intake & Tracking              | <ul style="list-style-type: none"> <li>• <b>New Case:</b> Submit individual cases with PDF files and optional signature references.</li> <li>• <b>Bulk Upload:</b> Upload a zip file containing multiple case folders.</li> <li>• <b>My Cases:</b> Track status (<i>Under review, Cleared, Action required</i>) of own submissions.</li> <li>• <b>Case Detail (Submitter View):</b> View uploaded documents and extracted fields without risk scoring details.</li> </ul>                                                                                                                                                                     |
| <b>Reviewer L1</b> ( <i>Analyst</i> )         | Initial Verification & Risk Triage      | <ul style="list-style-type: none"> <li>• <i>All User capabilities</i>, plus:</li> <li>• <b>Dashboard:</b> High-level case volume and risk distribution metrics.</li> <li>• <b>Review Queue:</b> Filter cases by risk tier, document category, and status.</li> <li>• <b>Full Case Detail:</b> View risk scores (0–100), triggered fraud reasons, and automated forensic checks.</li> <li>• <b>Visual Overlay Viewer:</b> Inspect bounding box anomalies directly on document pages.</li> <li>• <b>Decision Actions:</b> Approve, Reject, or Escalate cases.</li> <li>• <b>Export Report:</b> Download full forensic audit reports.</li> </ul> |
| <b>Reviewer L2</b> ( <i>Senior Reviewer</i> ) | Final Adjudication & Company Governance | <ul style="list-style-type: none"> <li>• <i>All Reviewer L1 capabilities</i>, plus:</li> <li>• <b>Escalated Cases Resolution:</b> Final approve/reject authority on complex cases escalated by L1 analysts.</li> <li>• <b>Issuer Registry:</b> Add, edit, verify, or deactivate trusted vendors and issuers for your company.</li> <li>• <b>Company Risk Rules:</b> Customize fraud rule weights and adjust Low/Medium/High risk tier score thresholds.</li> </ul>                                                                                                                                                                            |

## 3. Signing In & Navigation

Access the platform URL provided by your organization using your business email address and password.



FDDT Sign In Screen

1. Enter your corporate email address and assigned password.
2. Click **Sign in**.
3. Upon successful sign-in, the top navigation bar displays your organization name, user role badge, and relevant navigation links based on your permissions.

*Note: If you have forgotten your password or require a new user account, please email your administrator for this.*

**Changing your password.** Once signed in, click the **key icon** in the top navigation bar (or **Change password** in the mobile menu), enter your current password and the new one twice (at least 8 characters). Your current password is required — a forgotten password can only be reset by your administrator.

## 4. User (Submitter) Guide

### Submitting a Single Case

To submit documents for verification:

1. Click **New Case** (or **New upload**) in the top navigation bar.
2. Select the **Case Type** that matches your submission:
  - School / educational document
  - Vendor invoice
  - Commercial invoice
  - Procurement documentation
  - Quotation
  - Travel / accommodation reimbursement
  - Other

**FDDT**  
FRAUD DOCUMENT DETECTION TOOL

Dashboard Cases My cases

testcompany User New upload TU Sign out

**SECURE INTAKE**

## Analyze new documents

Upload related evidence together for cross-document verification.

**Submit new case**

Select a case type and attach every supporting document — they will all be linked to the same case for cross-document checks.

**Case type**

Vendor invoice

**Supporting documents**

Drag and drop files here, or click to browse

Attach every document supporting this case — e.g. an invoice and its approval voucher — in one go.  
PDF only · max file size: 10.0 MB each · not password-protected

**Analysis pipeline**

- ✓ Secure original & verify hash
- ✓ Extract layout and fields
- 3 Validate issuer and consistency
- 4 Run forensic checks
- 5 Calculate explainable risk

**EVIDENCE INTEGRITY**

**Originals remain unchanged**

Every uploaded file is preserved with a verifiable integrity hash and complete audit history.

New Case Intake Form

### 3. Attach PDF Files:

- Drag and drop your PDF documents into the upload zone or click **Browse files**.
- **File Requirements:** Files must be valid PDF documents within your company's maximum file size limit (standard limit is 10 MB per document). Password-protected PDFs are rejected immediately for security.

### 4. Optional Reference Signature:

- If your document contains an authorized signature or stamp that should be compared against other documents in the case, you can define a reference signature box.

5. Click **Submit Case**. The documents will begin automated processing immediately.

## Submitting Cases in Bulk (Zip Upload)

When managing high volumes of invoices or expense claims, use the **Bulk Upload** feature:

**FDDT**  
FRAUD DOCUMENT DETECTION TOOL

Dashboard Cases My cases

testcompany User New upload TU Sign out

**SECURE INTAKE**

## Bulk upload

Submit many cases at once: one zip file, one folder per case.

**Upload a zip of cases**

Each top-level folder in the zip becomes its own case, and every document directly inside that folder is attached to it. Cases are created and processed one by one, so results start appearing within seconds.

**Case type (applies to every case in the zip)**

Vendor invoice

**Zip file**

Drag and drop a .zip here, or click to browse

Max zip size: 300.0 MB · max file size: 10.0 MB per PDF

**How to structure the zip**

claims.zip

- case-001/ invoice.pdf, receipt.pdf
- case-002/ quotation.pdf
- case-003/ ...

- Put each case's documents directly in its folder. A case folder that contains a subfolder is rejected.
- The folder name becomes the case's reference label; it doesn't need to be unique.
- A bad file is rejected on its own; the rest of its case still goes ahead.
- Arabic and other non-English file and folder names are supported.

Bulk Upload Interface

1. Navigate to **New Case** and select **Bulk upload (zip of cases)**.

2. Prepare a standard `.zip` file on your computer structured with **one folder per case**:

```
batch_october.zip
├─ CASE_INV_1001/
│  └─ invoice_1001.pdf
│     └─ purchase_order.pdf
├─ CASE_INV_1002/
│  └─ invoice_1002.pdf
└─ CASE_INV_1003/
   └─ tuition_receipt.pdf
      └─ student_id.pdf
```

- 3. Select the default **Case Type** for the batch.
- 4. Drop the `.zip` archive into the intake area.
- 5. The system automatically extracts each folder, verifies the PDF files, and creates individual cases with automated progress indicators (**Queued** → **Processing** → **Complete**).

## Tracking Submissions (My Cases)

Submitters can track all their submitted cases under **My Cases**:

FDDT  
FRAUD DOCUMENT DETECTION TOOL

DashboardCasesMy cases

testcompany  
User

New uploadTU[→ Sign out]

SUBMISSION TRACKER

My Cases

Track the verification and clearance progress of your submitted document packages.

Export CSV

Submit new case

TOTAL SUBMITTED1

UNDER REVIEW0

CLEARED / VERIFIED0

ACTION REQUIRED1

Q Search by Case ID or filename...

Status: All StatusesReset

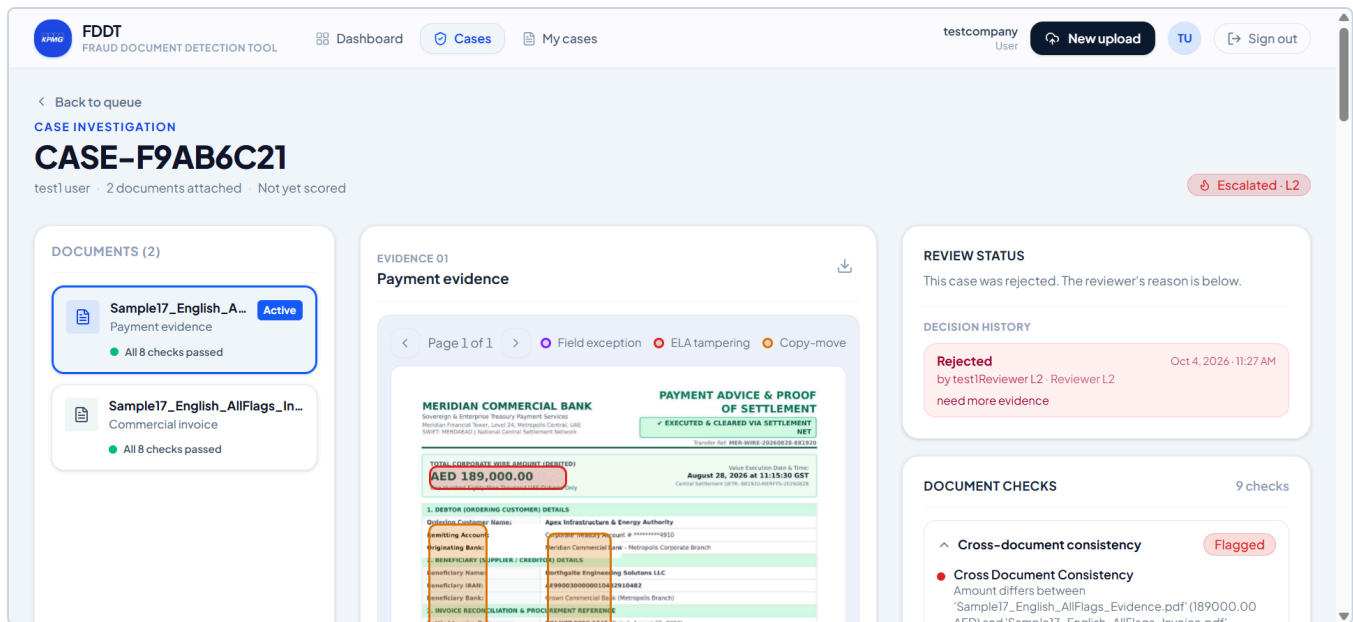
| CASE ID                            | TYPE               | DOCUMENTS | STATUS          | SUBMITTED              | ACTIONS        |
|------------------------------------|--------------------|-----------|-----------------|------------------------|----------------|
| CASE-F9AB6C21<br>CASE-F9AB6C21.pdf | Commercial invoice | 2 docs    | Action Required | Oct 4, 2026 - 11:18 AM | View details → |

My Cases Dashboard

- **Search:** Search by case reference number or document name.
- **Filters:**
  - **All:** All cases submitted by your user account.
  - **Under review:** Cases currently being analyzed by the automated pipeline or queued for reviewer triage.
  - **Cleared:** Cases approved by the reviewing team.
  - **Action required:** Cases rejected with feedback notes from the reviewer.

## Understanding the Submitter Case View

When a submitter opens one of their cases, a clean, submitter-tailored view is presented:



Submitter Case Detail View

- **Document Overview:** Lists all uploaded files, page counts, file sizes, and processing statuses.
- **Extracted Information:** Displays values recognized by the OCR engine (such as invoice number, total amount, currency, and date).
- **Activity Timeline:** Displays a chronological event log showing when the case was submitted, when files were processed, and final decisions.

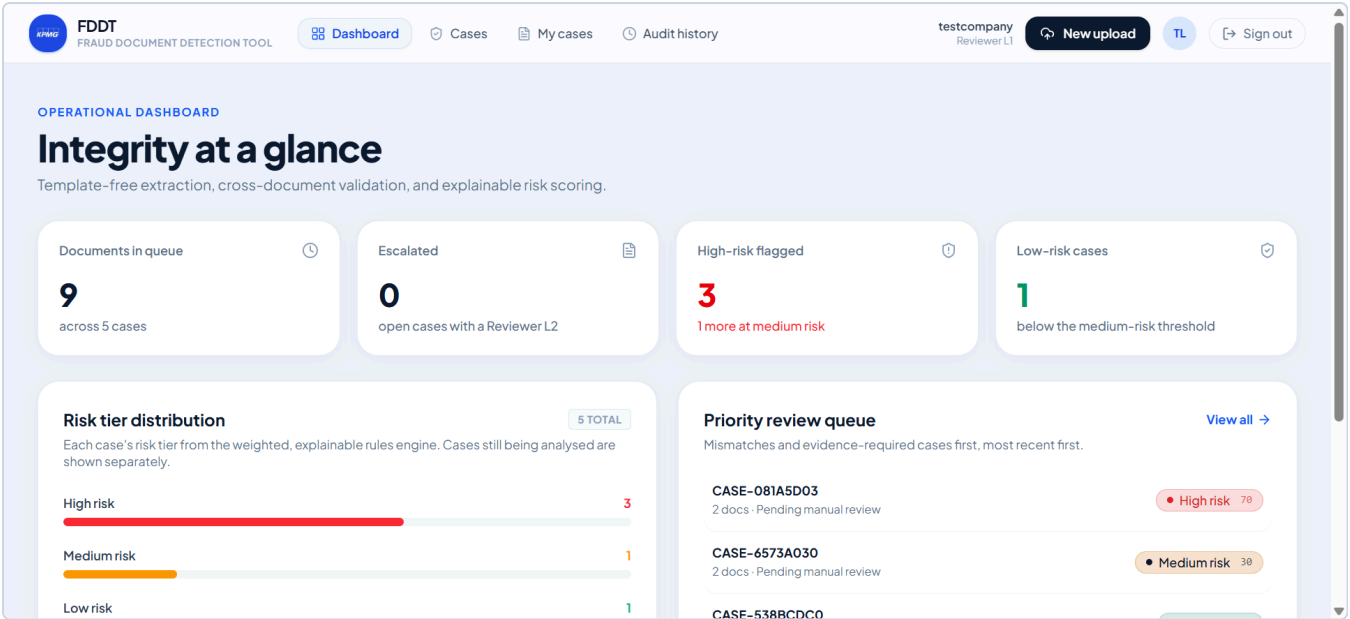
**Security & Fraud Prevention Design:** Submitter accounts do **not** display numerical risk scores, triggered fraud check rules, or forensic detection algorithms. This prevents malicious actors from probing or attempting to bypass verification thresholds.

## 5. Reviewer L1 (Analyst) Guide

Reviewer L1 analysts assess automated findings, inspect flagged anomalies, and make informed operational decisions.

### Operational Reviewer Dashboard

The **Dashboard** provides real-time visibility into your company's verification queue:

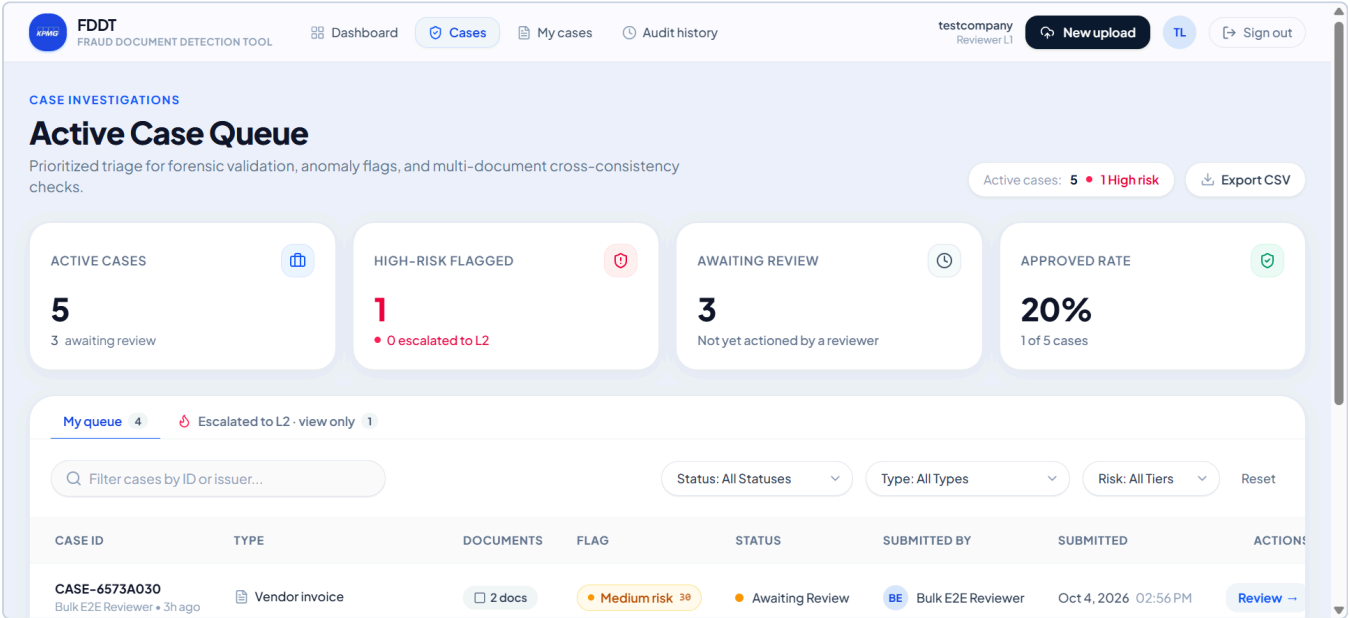


Reviewer Operational Dashboard

- **Key Metrics:** Total active cases, cases requiring immediate review, cleared cases, and flagged cases.
- **Risk Distribution:** High, Medium, and Low risk breakdowns across recent submissions.
- **Review Workload:** Cases sorted by submission date and SLA urgency.

Managing the Review Queue

Click **Cases** in the navigation bar to access the full **Review Queue**:



Review Queue

- **Risk Tier Badges:**
  - **HIGH RISK** (Score 70–100): Critical anomalies detected (e.g., severe metadata tampering, copy-move duplication, unrecognized issuer).
  - **MEDIUM RISK** (Score 30–69): Discrepancies or warnings (e.g., date mismatches, mathematical errors, missing expected fields).
  - **LOW RISK** (Score 0–29): Clear document with verified issuer and consistent fields.

- **Filters & Search:** Filter by Risk Tier, Case Type, Review Status, or keyword.

## Full Case Analysis & Risk Breakdown

Opening any case from the queue reveals the complete **Case Detail** analysis:

**DDT**  
FRAUD DOCUMENT DETECTION TOOL

Dashboard

Cases

My cases

Audit history

testcompany  
Reviewer L1

New upload

TL

Sign out

[Back to queue](#)

## CASE INVESTIGATION

# CASE-538BCDCO

Bulk E2E Reviewer · 1 document attached · Low

Export report

Low risk 15/100

Issuer "OAKRIDGE INTERNATIONAL ACADEMY" on "invoice.pdf" did not match any known issuer in the registry.

DOCUMENTS (1)

invoice.pdf

School / educational document

Active

All 8 checks passed

EVIDENCE 01

School / educational document

Page 1 of 1

OAKRIDGE INTERNATIONAL ACADEMY

TUITION FEE INVOICE

Excellence in Global Secondary Education | Accredited IB World School  
1200 Academy Boulevard, Boston, MA 02118  
School Tax ID: 06-303-0206 | Tel: +1 (857) 303-8900 | [tours@oakridge.edu](mailto:tours@oakridge.edu)

INVOICE #: OAK-2026-4910  
Issue Date: August 15, 2026  
Due Date: September 15, 2026  
Academic Term: Fall Semester 2026

| STUDENT INFORMATION                                     |  | PARENT / SPONSOR BILLING DETAILS                               |  |
|---------------------------------------------------------|--|----------------------------------------------------------------|--|
| <b>Student Name:</b> Alexander Hayes                    |  | <b>Parent / Employee Name:</b> David Hayes                     |  |
| <b>Student ID:</b> STU-2026-0912                        |  | <b>Corporate Employer:</b> Horizon Technologies Inc.           |  |
| <b>Enrolled Grades:</b> Grade 9 (High School Freshman)  |  | <b>Employee ID:</b> EMP-77492 (Education Allowance Claim)      |  |
| <b>Curriculum:</b> International Baccalaureate (IB MYP) |  | <b>Billing Address:</b> 42 Commonwealth Ave., Boston, MA 02118 |  |

| # | Tuition & Educational Fee Description                                                                                        | Period    | Rate (\$) | Total (\$) |
|---|------------------------------------------------------------------------------------------------------------------------------|-----------|-----------|------------|
| 1 | <b>Grade 9 Full Tuition (Term 1 - Fall 2026)</b><br>Core curriculum instruction, faculty salary, & academic support          | Fall 2026 | 11,200.00 | 11,200.00  |
| 2 | <b>STEM Science Lab &amp; Digital Technology Fee</b><br>High school subject laboratory access, software licenses & resources | Annual    | 1,650.00  | 1,650.00   |
| 3 | <b>School Transport &amp; Athletic Program Fee</b><br>Dedicated shuttle service route & competitive sports registration      | Term 1    | 1,400.00  | 1,400.00   |

Risk assessment

Low risk

15 / 100

Sum of the weights of 1 triggered rule, capped at 100. Tiers: medium from 30, high from 60 (as configured when scored). Advisory – for reviewer use.

REVIEWER DECISION

Approve

Reject

Escalate

Escalating hands the case to a Reviewer L2; Reviewer L1s keep read-only access.

## Reviewer Case Overview

### 1. Risk Score Header:

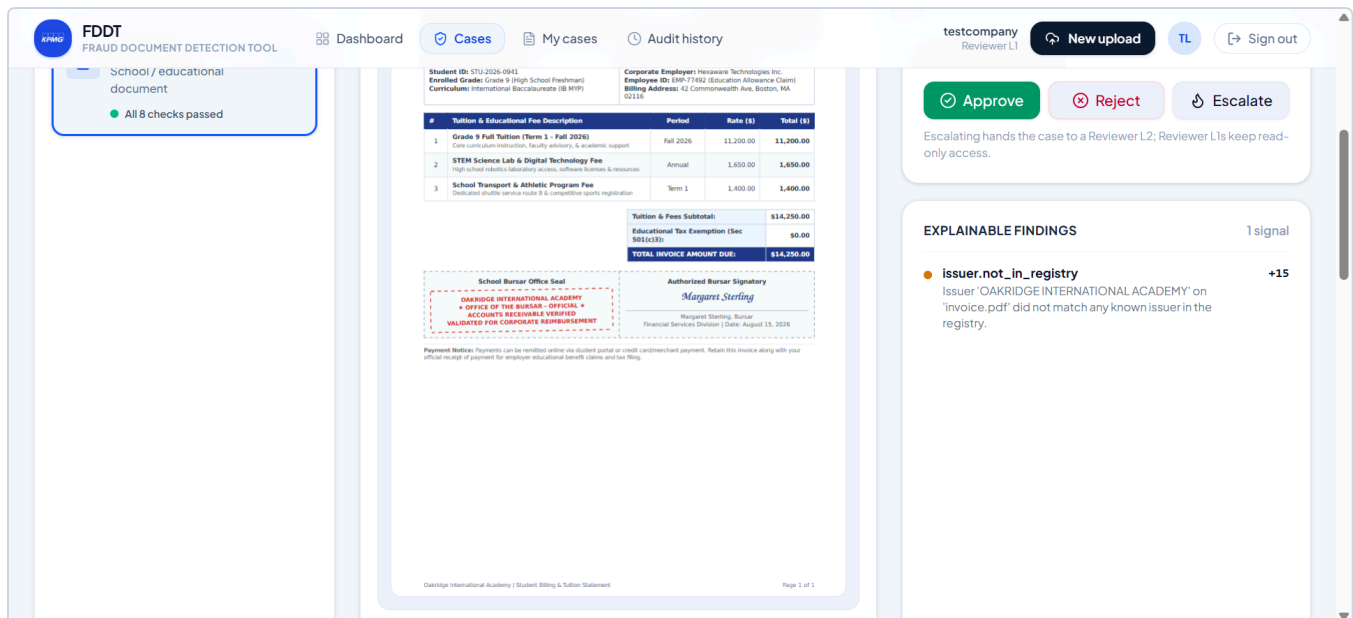
- Prominently displays the composite Risk Score (0–100) and assigned Tier.
- Shows case metadata: submitter email, case category, submission timestamp.

## 2. Triggered Risk Reasons:

- Clear, explainable list of factors that contributed points to the score.
- Example: "*Issuer not found in verified registry (+25 pts)*", "*PDF modified in image editor after creation (+20 pts)*".

## Forensic & Automated Document Checks

Scroll down on the Case Detail page to inspect the comprehensive **Checks & Forensics Panel**:



Automated Checks & Forensics Panel

Every uploaded document undergoes an exhaustive battery of independent tests:

| Check Name                         | What It Inspects                                                                                                       |
|------------------------------------|------------------------------------------------------------------------------------------------------------------------|
| <b>Issuer Verification</b>         | Cross-references extracted vendor name, tax ID, and address against your company's trusted Issuer Registry.            |
| <b>Field Validation</b>            | Verifies arithmetic totals (subtotal + tax = total), valid date formats, future dates, and required fields.            |
| <b>Cross-Document Check</b>        | Validates consistency between multiple documents in the same case (e.g., matching invoice amount with purchase order). |
| <b>Metadata Forensics</b>          | Inspects PDF creation software, modification history, suspicious editing tools, and timestamp inconsistencies.         |
| <b>Copy-Move Forgery</b>           | Detects cloned regions (content copied and pasted from one part of the document to another).                           |
| <b>Error Level Analysis (ELA)</b>  | Analyzes compression resave artifacts to identify spliced text or altered numbers.                                     |
| <b>Duplicate Detection</b>         | Computes cryptographic and perceptual hashes to detect duplicate submissions across historical cases.                  |
| <b>Signature / Stamp Detection</b> | Detects signature regions and compares them against reference samples for consistency.                                 |

## Interactive PDF Visual Overlays

To visually inspect anomalies:

- Open the document preview within the Case Detail view.
- Color-coded bounding boxes highlight exact regions of concern:
  - **Red Box:** Tampering or Error Level Analysis anomaly.
  - **Orange Box:** Copy-move duplication source and target.
  - **Purple Box:** Arithmetic or field validation mismatch.
  - **Fuchsia Box:** Text set in a different font (or a second copy of the font) — likely typed in later.
  - **Teal Box:** Deleted or shortened text — the faint trace of erased text in a converted scan.

- **Blue Box:** Detected signature or official stamp.

## Review Decisions (Approve, Reject, Escalate)

In the top-right header of the Case Detail screen, three decision actions are available:

The screenshot displays the FDDT (Fraud Document Detection Tool) interface for Case-538BCDC0. The top navigation bar includes 'Dashboard', 'Cases', 'My cases', and 'Audit history'. The case status is 'Low risk' with a score of 15/100. A document 'invoice.pdf' is attached, showing 'All 8 checks passed'. The central area displays the 'Tuition Fee Invoice' from Oakridge International Academy, detailing student information, parent details, and a table of fees. The 'Reviewer Decision' section shows the case is 'Approved' by testreviewer L1 on Oct 4, 2026.

Reviewer Decision Modal

### 1. Approve:

- Use when all checks pass or minor flags have been verified as legitimate.
- Enter optional approval notes and confirm. The case status updates to **Approved**.

### 2. Reject:

- Use when fraudulent alteration or non-compliance is identified.
- Select a mandatory **Rejection Reason** category and provide notes detailing the decision.

### 3. Escalate:

- Use when a case is complex, high-value, or requires supervisor adjudication.
- Adds an escalation note and routes the case directly to the **Reviewer L2** queue.

## Exporting Case Forensic Reports

Click **Export report** at any time on the Case Detail page to generate an official, courtroom-ready **PDF Forensic Report**:

- Includes complete executive summary, breakdown of all automated checks, risk scoring explanation, audit trail, SHA-256 cryptographic document hashes, and high-resolution visual evidence pages with highlighted anomaly boxes.

## 6. Reviewer L2 (Senior Reviewer / Supervisor) Guide

Reviewer L2 users have all analyst privileges plus senior decision authority and configuration control over their company's risk parameters.

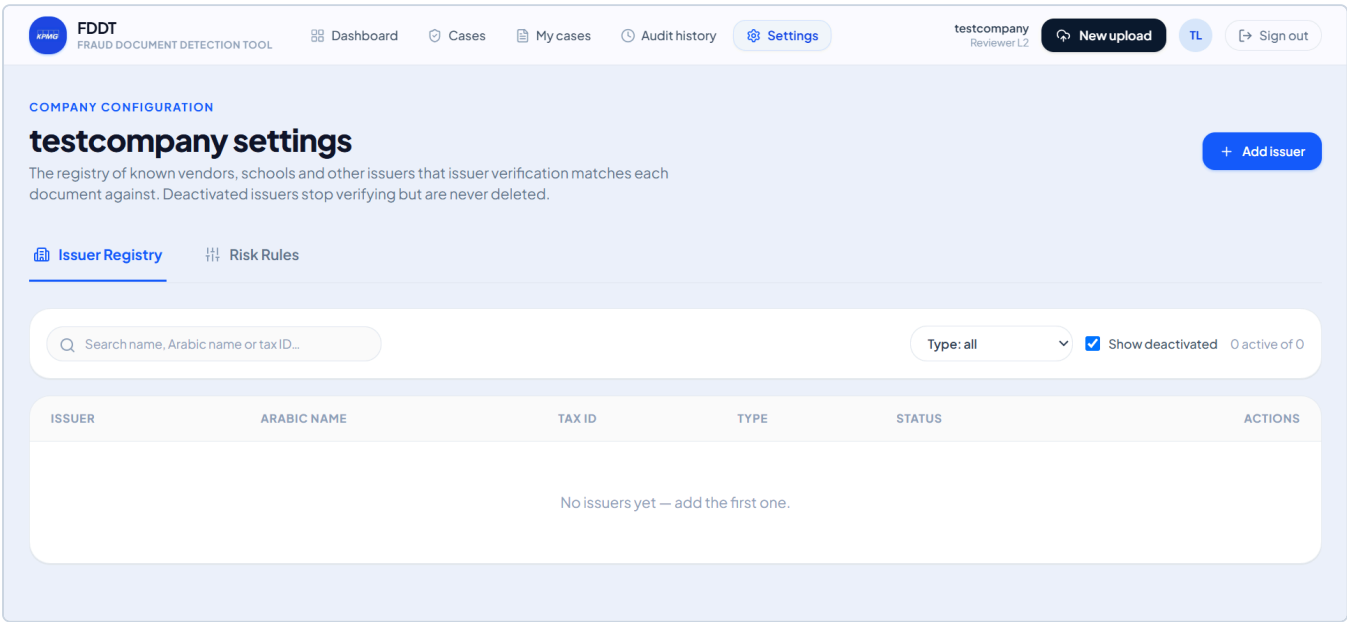
## Managing Escalated Cases

Cases escalated by L1 analysts appear in the review queue with an **Escalated** status badge.

- Reviewer L2 users conduct deep reviews of these cases.
- Only Reviewer L2 users have the authority to grant final **Approval** or execute final **Rejection** on escalated files.

## Company Issuer Registry

Under **Settings > Issuer Registry**, Reviewer L2 supervisors manage the company's verified vendors, schools, and partners:

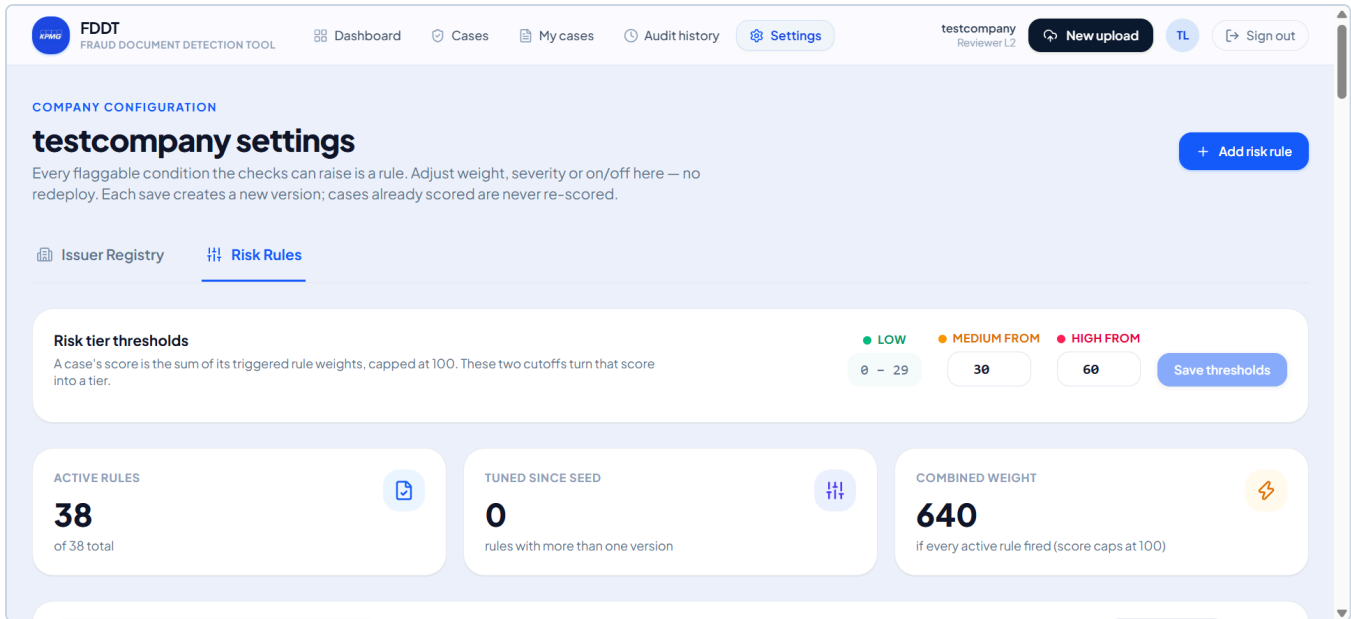


Issuer Registry Management

1. **Viewing Registry:** Lists all active and inactive registered issuers.
2. **Adding an Issuer:**
  - Click **Add Issuer**.
  - Enter Organization Name, Tax ID / Business Registration Number, official address, and known email domains.
  - Save the record. Incoming documents matching these details will pass Issuer Verification automatically.
3. **Deactivating / Reactivating:**
  - Deactivate obsolete vendors or reactivate previously deactivated entities with a single click.

## Company Risk Rules & Thresholds

Under **Settings > Risk Rules**, Reviewer L2 supervisors customize how risk scores are calculated for their organization:



Company Risk Rules Management

### 1. Adjusting Risk Rules:

- Toggle individual fraud detection checks on or off based on your business requirements.
- Adjust point weights for specific findings (e.g., increasing penalty points for metadata anomalies).

### 2. Configuring Risk Thresholds:

- Customize score boundaries for **Low**, **Medium**, and **High** risk classifications to match your company's risk tolerance.

## 7. Getting Help & Administrative Requests

To maintain company security and data integrity, certain platform-level configurations are managed centrally.

### Need assistance?

- To create new user accounts or adjust user roles
- To reset an account password
- To request an increase in maximum document upload size limits
- To update your company profile or contact details

**Please contact or email your administrator at [admin@example.com](mailto:admin@example.com) (or your company's designated IT administrator) for this.**